

Context-Dependent Selection Among Established Urban Routing Policies: Complementarity, Decision Value and Generalisation Under Distribution Shift

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Abstract

A route guidance system must commit to a routing objective before any trip is assigned, and the merit of that objective depends on the operating context. We study the selection of the routing *policy* itself as a per-instance decision problem. Four established policies — shortest distance, reactive travel time, capacity-aware load balancing and reliability-aware mean–variance guidance — are evaluated on a 648-context full factorial over corridor demand, green ratio, guidance penetration, information lag, disruption and alternative capacity, with five paired seeds per cell: 12,960 valid microsimulation runs. Three of the four policies are strictly best somewhere with a seed-resolved margin, and 73.0 % of contexts have more than one non-dominated policy across journey time, CO₂ and stopped delay, so the portfolio is complementary. Its decision value is small: the best fixed policy leaves 0.45 s per vehicle of headroom on a 474.4 s mean journey time, whereas the worst fixed policy costs 220.2 s (46.4 %). A gradient-boosted selector trained on advantage rather than on the winner’s identity recovers 55.8 % of that headroom under leave-one-context-out evaluation and halves the worst case, while a transparent mechanistic rule is *more* accurate and *more* costly than doing nothing. Across eight distribution shifts reported separately by type the selector helps on two, is indistinguishable on three and harms on three; a support-and-confidence gate never certified a deviation, because the conformal interval is two orders of magnitude wider than the effect, and flagged none of the domain shift its features cannot encode. The contribution is a measurement rather than an algorithm: complementarity, decision value and deployability are distinct and quantitatively disconnected here, so reporting the first as evidence for the third overstates the case for adaptive routing-policy selection. Results are

scoped to one synthetic network.

Keywords: routing-policy selection, algorithm selection, decision regret, distribution shift, selective prediction, traffic microsimulation

1. Introduction

Every deployed route guidance system embeds an answer to a question it is rarely asked to justify: by what objective should its paths be computed? Minimising distance, tracking the most recently observed travel times, balancing predicted flow against link capacity and penalising the variance of travel time are all defensible answers, and they are not the same answer — each sends a given request down a different corridor. The objective is therefore a decision variable in its own right, logically prior to and constraining the route computation that follows it. It is that prior decision, rather than the route computation, that this paper measures.

The question is operational rather than algorithmic. An operator does not usually get to invent a routing mechanism; it gets to choose among established ones, and to change that choice as conditions change. Whether it should is not obvious. The relative merit of routing objectives is known to depend on the traffic state: reacting to observed travel times helps most when there is congestion to react to, and can hurt when there is not, because a policy that diverts traffic from the shortest corridor pays the diversion cost whether or not the corridor is loaded. Modern systems also expose far more of the operating context than demand alone. Guidance reaches a varying and often unknown share of vehicles; the information it acts on is delayed by a measurable lag; signal timing determines how much of the available capacity each movement receives; capacity itself changes with disruption. Each of these is a plausible reason for the preferred policy to change, and each is measurable before any vehicle departs.

Treating an operating condition as a problem instance and a routing objective as a solver places the question inside the per-instance algorithm-selection framework [15, 3], which supplies both the baselines and the vocabulary used here. The reason for adopting that framing is not convenience but the separation it forces. Whether some policies beat others in some conditions, how much money that variation is actually worth, and whether any rule can collect it from information available in advance are three questions with three different answers, measured in three different units — a count of contexts,

a cost per vehicle, and a held-out regret under shift. They are routinely reported as though the first implied the last. Work on routing objectives typically establishes the first and leaves the other two to inference, which is safe only when the gap between them is small. Measuring that gap is the purpose of this study.

We therefore ask:

- RQ1** Under which combinations of demand, signal timing, guidance penetration, information lag, disruption and alternative capacity do established routing policies exhibit complementary strengths?
- RQ2** Which physical properties of the context explain where the preferred policy changes?
- RQ3** How much decision value is available from context-dependent selection, relative to the strongest fixed policy and to the hindsight bound?
- RQ4** Does a learned selector contribute anything beyond a transparent mechanistic rule, and by what measure?
- RQ5** How does selector performance change under interpolation, extrapolation, concept shift and domain shift, and what can an abstention mechanism actually prevent?

The study answers these on a controlled microsimulation benchmark: a three-corridor urban network, four established policies, a 648-cell full factorial over the six factors above, five paired seeds per cell, and 12,960 valid runs with no failures and no vehicle teleports. The contributions are empirical and methodological, and one of them is negative:

1. A controlled characterisation of routing-policy complementarity across six operating dimensions simultaneously, with an explicit account of ties and of which wins survive replication noise (Section 6.2).
2. A quantitative separation of complementarity from decision value. The portfolio is complementary on three of four policies and on the criterion vector in 73.0% of contexts, yet the headroom over the best fixed policy is 0.095% of mean journey time, while the penalty for the wrong fixed policy is 46.4% (Section 6.4).

3. Evidence that decision-oriented evaluation reverses the ordering that accuracy would give. A depth-2 mechanistic rule is more accurate than doing nothing and costs more; a gradient-boosted model trained on advantage is less accurate than a logistic classifier and costs less (Section 6.5).
4. A distribution-shift analysis reported separately by shift type, in which the selector helps on two splits, is indistinguishable on three and harms on three, and in which the support gate flags every context of three extrapolation splits but none of the domain shift it was intended for (Section 6.7).
5. A negative result about selective prediction in this regime: the confidence gate never certified a deviation in any context, in or out of distribution, because the conformal half-width exceeds the available advantage by two orders of magnitude (Section 6.8).

Nothing here is proposed as a new method. Each policy is taken from the literature unchanged, criteria are compared by dominance rather than through a new preference model, and the selectors are standard classifiers and regressors used off the shelf. All results belong to one synthetic environment and none has been checked against field data. Section 2 positions the study, Section 3 states the decision problem, Sections 4 and 5 describe the method and the computational study, Section 6 reports results, Section 7 interprets them, and Section 8 concludes.

2. Literature Review and Research Gap

Three bodies of work bear on the problem. The first supplies the routing mechanisms and what is known about when they help. The second supplies the objectives beyond travel time and the machinery for comparing them. The third supplies the decision-theoretic apparatus for choosing among mechanisms and for judging whether the choice generalises. Each is summarised for what it establishes; Section 2.4 states what none of them settles.

2.1. *Dynamic and Traffic-Aware Routing*

Route guidance begins from the observation that individually optimal and system-optimal assignments differ [23]. Dynamic guidance replaces static link attributes with observed or predicted conditions, and proactive formulations spread vehicles over admissible alternatives rather than concentrating them

on the current fastest path; the flow-balanced k -shortest path construction of Pan et al. [14] is the direct ancestor of the capacity-aware policy used here.

A second and more cautionary strand establishes that guidance is not monotonically beneficial. Providing travellers with information need not reduce congestion and can increase it, both in equilibrium models [2] and in networks where added capacity or added information degrades outcomes [1]. The size and even the sign of the effect depends on how many vehicles are guided: Yang [25] shows that benefits vary non-monotonically with market penetration, and recent work reaches the same conclusion from other directions. Toso et al. [21] show in a dynamical network-flow model that recommendation-following can produce partial demand transfer and congestion build-up whose severity is governed by the penetration rate, and Cornacchia et al. [6], simulating commercial navigation services on three Italian cities, find that route diversity falls as adoption rises and that emission benefits diminish, vanish or reverse beyond a city- and service-specific adoption threshold.

This literature therefore establishes that a routing policy’s value is state-dependent and penetration-dependent, and that the dependence can change sign. What it does not do is treat the choice among established policies as the object of study: it compares mechanisms, or studies one mechanism as adoption varies, rather than asking which of several should be active in a given operating condition and what that choice is worth.

2.2. Environmental, Multi-Criteria and Context-Dependent Routing

Routing on criteria other than time is well established. Eco-routing assigns paths using modelled emissions or fuel consumption [4], and microsimulation supplies per-vehicle emission models that make such criteria computable inside a simulation [11]. Reliability-aware guidance treats the variance of travel time as a cost alongside its mean; the mean–variance formulation of Sen et al. [16] is the ancestor of the reliability-aware policy used here.

Where several criteria are in play, the question of which policy is “preferred” becomes a question about the criterion vector rather than about a scalar. That distinction matters here for a specific reason: a policy can be non-dominated on the criterion vector while never being best on any single criterion, and a study that scalarises early will not see it. This study therefore reports Pareto structure directly, and uses scalarisation only as a declared sensitivity analysis. It does not propose a multi-criteria decision method, and it deliberately avoids

monetary scalarisation, because no value of time, value of reliability or carbon price was available for this setting that could be sourced rather than assumed.

2.3. Algorithm Selection and Decision-Oriented Evaluation

The selection problem was formalised by Rice [15] as a mapping from measurable properties of a problem instance to the algorithm expected to solve it best. Solver portfolios put that mapping into practice [24], benchmark libraries fix how the resulting systems are described and compared [3], and surveys together with instance-space analysis set out the methodology and where it breaks [10, 19, 18]. Two reference points from that literature are used throughout this paper: the single best solver, the strongest deployable fixed choice, and the virtual best solver, the per-instance best chosen with hindsight, which is a bound and not a method.

The same literature supplies the argument for evaluating a selector by the cost of its decisions rather than by its predictive accuracy. A model may predict outcomes poorly and still decide well if it preserves the ordering of the alternatives, and may predict well and decide badly if a small error falls near a boundary [7]. Where a selector must be trusted outside the conditions it was fitted on, the relevant machinery is that of distribution shift and selective prediction: the failure of exchangeability under covariate shift [17], conformal prediction and its weighted extension [22, 12, 20], and the classical reject option [5]. Gradient boosting supplies the regression model used here [8, 9].

2.4. Research Gap and Positioning

Each strand answers part of the question and none answers the whole of it. Traffic-aware routing research demonstrates state dependence but keeps the mechanism, not the choice between mechanisms, as its object. Multi-criteria routing research supplies objectives and the means of comparing them without asking whether, in a particular network, those objectives separate the candidates at all. Algorithm-selection research supplies baselines, regret and generalisation testing, but its traffic applications are scarce and none pairs a controlled factorial over operating conditions with a shift analysis reported by shift type.

The gap is accordingly not a claim that regime-dependent routing is unstudied; it plainly is not. The gap is that the three quantities named at the end of Section 1 have not been measured on a common benchmark, so their relationship is unknown, and a reader has no basis for converting evidence of the first into an expectation about the third. Every ingredient used below

has precedent and is cited above. The contribution is the joint measurement, on a design large enough to separate the three quantities from one another and from the replication noise of the simulator, and the reporting of what that measurement returns, including where it returns nothing.

3. Decision Problem and Conceptual Framework

3.1. Policy Selection Versus Route Assignment

Write $\mathcal{S}$ for the space of operating contexts and $\mathcal{A}$ for the finite portfolio of routing policies. Two nested decisions are made. The outer one, which we call *level 1*, observes a context $s \in \mathcal{S}$ and switches on a single policy $a \in \mathcal{A}$ for the episode. The inner one, *level 2*, is carried out by that policy: it assigns a path to each guided vehicle from whatever traffic information it is entitled to read. Learning occurs only at level 1. The policies themselves are fixed constructions taken from the literature and are not altered, tuned or combined by the selector.

The separation matters for what can be claimed. Because level 2 is unchanged, an improvement attributable to level 1 is an improvement in *choosing*, not in routing, and it is bounded above by the best that the fixed portfolio can do. It also fixes the unit of analysis: a context is one instance, not one vehicle and not one replication, and every replication of a context moves together through every evaluation protocol.

A context contains only quantities defined before the policy is activated. Here it is the six-tuple of corridor demand, effective green ratio, guidance penetration, information lag, presence of a disruption and alternative-corridor capacity. No realised travel time, queue, flow or emission enters it.

3.2. Context Space and Policy Portfolio

The context space is the full factorial of the six factors, and the portfolio contains four established policies, described in Sections 4.1 and 4.2. Two properties of the portfolio are treated as empirical questions rather than assumptions: whether its members are behaviourally distinct in this environment, and whether their outcome differences survive replication noise. Both are reported.

3.3. Decision Cost and Regret

Let $C(s, a)$ be the decision cost of policy a in context s , measured on the primary criterion and averaged over the replications of that context. The

primary criterion is the system mean journey time in seconds per vehicle; Section 4.4 states why, and the other two criteria are carried alongside. Costs are reported in seconds throughout, so no normalisation convention can affect them.

The per-context best and its cost are

$$a^{\text{VB}}(s) \in \arg \min_{a \in \mathcal{A}} C(s, a), \quad C^{\text{VB}}(s) = \min_{a \in \mathcal{A}} C(s, a). \quad (1)$$

Selecting $a^{\text{VB}}(s)$ everywhere defines the *virtual best* reference. It uses the realised performance of every candidate and is therefore a bound, not a deployable method. For a selection rule $\pi : \mathcal{S} \rightarrow \mathcal{A}$ the *decision regret* in context s is

$$R(s, \pi) = C(s, \pi(s)) - C^{\text{VB}}(s) \geq 0. \quad (2)$$

Regret measures the cost of the decision, in seconds per vehicle. It is not a prediction error, and the two are not monotonically related.

The strongest deployable non-adaptive rule is the *single best* fixed policy

$$a^{\text{SB}} = \arg \min_{a \in \mathcal{A}} \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} C(s, a), \quad (3)$$

and the quantity that adaptation competes for is the *decision headroom*

$$H = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left(C(s, a^{\text{SB}}) - C^{\text{VB}}(s) \right), \quad G(\pi) = 1 - \frac{\overline{R(\pi)}}{H}, \quad (4)$$

where $G(\pi)$ is the share of the headroom a rule recovers: $G = 1$ at the virtual best, $G = 0$ at the single best fixed policy and $G < 0$ for a rule that is worse than doing nothing adaptive. In every held-out protocol, a^{SB} is identified from training contexts only, so the baseline is itself a fitted quantity and not an oracle.

3.4. Complementarity, Decision Value and Deployability

The framework separates three properties that are often treated as one.

Complementarity. Different policies are best in different contexts: $a^{\text{VB}}(s)$ is not constant over $\mathcal{S}$. This is a property of the portfolio and the environment alone. It can be asserted on a scalar criterion or on the criterion vector, and the two can disagree: a policy may be non-dominated in many contexts while never minimising any single criterion. Complementarity says nothing about magnitude.

Decision value. The cost carried by that variation, H in Eq. (4). The two properties come apart whenever the fixed policy is merely *narrowly* beaten in the contexts it loses: many distinct winners can then coexist with a vanishing H . Complementarity is thus a precondition for decision value and no more than that, and the two are not even commensurable — one counts contexts, the other prices seconds per vehicle.

Deployability. Whether any rule can actually collect H using only what is knowable before the policy is switched on, in conditions it was not fitted to, and whether it keeps doing so once those conditions drift. Formally this is $G(\pi)$ measured out of sample, together with how π behaves when the context distribution is deliberately moved. Decision value does not buy this: interpolating within a sampled region is a different act from extrapolating beyond it, and the second can cost more than it returns.

A fourth quantity is needed to interpret the other three. Because every context is replicated, each has a *replication noise band*: the dispersion of the paired seed differences between policies. A headroom smaller than that band is not merely uncertain, it is unobservable at the achieved sample size, and the distinction between a small real effect and an unresolved one must be reported rather than absorbed into an average. Section 4.7 states the criterion used, and Section 6.4 reports headroom against noise context by context.

4. Methodology

4.1. Simulation Environment and Network

Experiments use SUMO 1.27.1 with a one-second step and vehicle teleports disabled [13], so a vehicle that cannot enter the network waits and remains in the accounting rather than being removed from it. Emissions use the HBEFA3 PC_G_EU4 model [11] over a homogeneous fleet.

The network (Fig. 1a) is a single origin–destination corridor served by three parallel routes between a diverge and a merge. The central arterial C is the shortest (3000 m, two lanes, four signalised junctions); the north street N is longer and slower (3242 m, one or two lanes, four signalised junctions); the south bypass S is longest but uninterrupted (3963 m, one lane, unsignalised). Because every junction permits straight-through movements only, the set of loopless paths from the origin portal to the destination portal is exactly $\{C, N, S\}$. The candidate catalogue is therefore complete rather than a k -shortest-path approximation, which removes one degree of freedom from

the comparison: no policy can be advantaged by a better path-generation heuristic.

The three routes differ on the dimensions that make policy choice non-trivial: C has the highest capacity but is signal-interrupted and is the target of every distance-minimising assignment; N adds capacity only when it has two lanes; S has fixed capacity and no signal variance, so it is the reliable option. Corridor capacities were measured rather than assumed, by saturation-flow calibration runs reported with the scenario. Cross-street background demand is held fixed so that it does not confound the factorial. The three corridors share only the access and egress links, so a route-overlap index carries no information and is not used.

Each run simulates a warm-up followed by a measurement window; the measured cohort is every vehicle, guided or not, corridor or cross-street, with intended departure in the window. Using the system-wide cohort rather than the guided cohort is deliberate: a policy that improves guided vehicles by displacing congestion onto others should not be credited for it.

4.2. Routing Policies

All four policies choose one element of the three-path catalogue for one vehicle at its moment of insertion. Three of them read traffic state, and they read it from one shared observation buffer, so no policy has an information advantage over another. The buffer stores the instantaneous travel time of every corridor edge at a 30 s sampling interval. A policy acting at time t may read only samples with timestamp at most $t - \ell$, where ℓ is the context’s information lag, and only the most recent 120 s of those. Before any admissible sample exists it falls back to free-flow times. No policy sees the future, and none sees any realised outcome of the run it belongs to. Table 1 states each policy; the parameters were frozen before any evaluation run and are varied only in the sensitivity experiment of Section 5.4.

Eco-routing is not in the portfolio. It was admitted only conditionally, on the screening evidence of genuine decoupling between travel time and CO₂; that evidence did not appear (Section 5.1), and the exclusion is therefore a reported finding rather than a design assumption.

4.3. Experimental Factors

Six factors define a context (Fig. 1b). Corridor demand takes six levels from 1200 to 4200 veh/h, spanning free flow to oversaturation of the shortest corridor. The effective green ratio takes three levels, which changes the

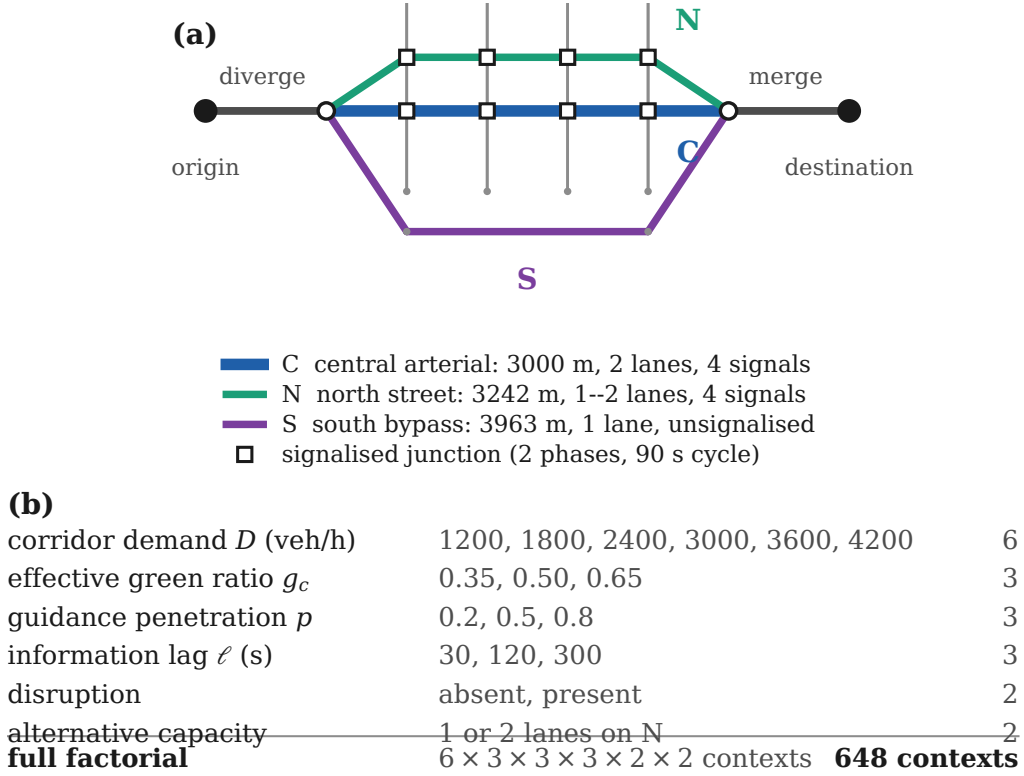


Figure 1: (a) The network. A single origin–destination corridor is served by three parallel routes between the diverge and the merge: the short signalised arterial C, the longer signalised street N, and the long unsignalised bypass S. Cross streets carry fixed background demand. (b) The experimental factors and their levels. The full factorial gives 648 contexts, each simulated under four policies with five paired seeds.

Table 1: The four routing policies. $L(p)$ is path length; $\hat{\mu}_p$ and $\hat{\sigma}_p$ are the mean and standard deviation of the path travel time over the visible observation window; u_e is the utilisation of link e .

	Objective and rule	Information used, parameters, and intended behaviour
P1	Shortest distance: $\arg \min_p L(p)$	No traffic state at all. The zero-information reference, and the only policy whose choice does not vary within a context. It loads the shortest corridor whether or not that corridor has capacity.
P2	Reactive travel time: $\arg \min_p \hat{\mu}_p(t - \ell)$	Lagged link travel times. Reactive rather than predictive: it extrapolates nothing, so the estimate it acts on is ℓ seconds old. Its hypothesised failure mode is moving the whole guided cohort together onto a corridor that was best one lag ago.
P3	Capacity-aware load balancing: among paths with $\hat{\mu}_p \leq (1 + \epsilon) \min_q \hat{\mu}_q$, take $\arg \min_p \max_{e \in p} u_e$	Lagged travel times and flows, plus the policy's own recent assignments that have not yet cleared a link. Follows the flow-balanced k -shortest-path family [14]. Frozen $\epsilon = 0.20$, assignment window 120 s. It spreads load when capacity is uneven and wastes time when there is no capacity shortage.
P4	Reliability-aware mean-variance: $\arg \min_p [\hat{\mu}_p + \lambda \hat{\sigma}_p]$	Lagged link travel times. $\hat{\sigma}_p$ is the dispersion of the <i>path</i> travel time across sample instants, so edge correlations within a path are preserved and the dispersion is observed rather than modelled. Frozen $\lambda = 1.0$. Follows mean-variance route guidance [16]; it avoids volatile corridors and is over-conservative when variance carries no risk.

capacity of both signalised corridors without changing the network. Guidance penetration takes three levels, which changes the size of the cohort a policy actually controls, and is the factor most directly implicated in the herding results of Section 2.1. Information lag takes three levels from 30 to 300 s, which is the staleness of the state a reactive policy acts on. A disruption regime is present or absent; when present, one lane closure on the arterial is drawn from a declared distribution — edge uniform over the three arterial links, onset uniform on $[600, 1200]$ s, duration uniform over $\{300, 600, 900\}$ s — from a stream seeded by the run seed alone, so the realisation is a property of the context and seed and never of the policy. Alternative capacity is one or two lanes on the north street, which determines whether load balancing has anything to balance. The full factorial is $6 \times 3 \times 3 \times 3 \times 2 \times 2 = 648$ contexts.

4.4. Performance Criteria

Three criteria are recorded for every run. C_1 is the mean journey time over the measured cohort, in seconds per vehicle, and includes insertion delay: a policy that oversaturates the network produces vehicles that cannot enter it, and excluding their wait would credit the policy for the queue it caused. C_2 is total tailpipe CO_2 over the cohort. C_3 is total stopped delay, the time vehicles spend below 0.1 m/s, which is what queueing costs in a form that neither of the other two isolates.

C_1 is the primary decision cost. It is chosen because it requires no weights and is the quantity a guidance system is usually asked to reduce; using it does not assert that the other two are unimportant, and Section 6.6 reports how often they disagree with it. The three criteria are compared by Pareto dominance rather than scalarised, and scalarisation appears only as a declared sensitivity analysis over three normalised preference profiles. No monetary value of time, value of reliability or carbon price is used, because no verified source for any of them was available for this setting and inventing one would make the resulting trade-off an artefact of the assumption.

4.5. Selector Models

A selector maps a context to a policy using eight dimensionless descriptors of the context, listed in Table 2. Each is a function of the context definition alone; none can be computed from any outcome, and the disruption descriptor encodes that a disruption *may* occur, never its realisation. The descriptors are chosen to be mechanistically interpretable rather than maximally predictive, so that a rule expressed in them has a traffic reading.

Table 2: The eight context descriptors. D is corridor demand, p penetration, ℓ information lag, g the effective green ratio, and cap_C , cap_N , cap_S the corridor capacities implied by g and the lane configuration. T_C^{ff} is the free-flow time on the arterial.

	Definition	Reading
x_1	$D/(\text{cap}_C + \text{cap}_N + \text{cap}_S)$	network saturation
x_2	D/cap_C	saturation of the shortest corridor if it took all demand
x_3	$\text{cap}_N/(\text{cap}_C + \text{cap}_N + \text{cap}_S)$	share of capacity on the alternative
x_4	$(g T_{\text{cyc}} - y)/T_{\text{cyc}}$	effective green ratio
x_5	p	guidance penetration
x_6	ℓ/T_C^{ff}	information staleness relative to a free-flow trip
x_7	disruption indicator	whether a disruption may occur
x_8	$p D/\text{cap}_C$	controlled-flow saturation: the share of the shortest corridor’s capacity that the guided cohort alone would consume

Five selectors are evaluated on a common footing, together with the two references of Section 3.3:

- B0, fixed policy.** The single best fixed policy of Eq. (3), identified on the training contexts of each fold. This is the baseline adaptation must beat, and it is a fitted quantity.
- B1, mechanistic rule.** A depth-2 decision tree on the eight descriptors, fitted to the identity of the per-context winner. It is included because it is the kind of rule an engineer would write, and because its splits are readable as traffic statements.
- B2, deeper tree.** The same construction at depth 3.
- B3, logistic.** Multinomial logistic regression on the standardised descriptors, again fitted to the winner’s identity.
- B4, advantage GBDT.** Gradient-boosted regression [8, 9], but trained on a different target: for each policy a it predicts the *advantage* $\Delta_a(s) = C(s, a^{\text{SB}}) - C(s, a)$ over the training-fold fixed policy, and selects $\arg \max_a \hat{\Delta}_a$. This is the predict-then-optimize construction [7]:

the model is fitted to decision cost rather than to the winner’s label, so an error that does not change the ordering costs nothing.

B5, selective. B4 with two gates. A *support* gate flags a context whose mean distance to its five nearest training contexts, in the standardised descriptor space, exceeds the 95th percentile of that statistic within the training fold. A *confidence* gate requires the split-conformal lower bound on the predicted advantage, at miscoverage $\alpha = 0.10$ calibrated on a held-out 30 % of the training fold, to be positive. The selector acts only if both gates pass; otherwise it abstains to B0. This is a reject option [5] built on split conformal prediction [22, 12].

All hyperparameters were fixed in the frozen specification before any evaluation run and are not tuned. No neural, graph, reinforcement-learning or bandit model is used: the question is whether a flexible model adds anything over a transparent rule, not which flexible model wins.

4.6. *Distribution-Shift Framework*

Eight shift splits are evaluated, each declared before the analysis and each reported on its own. Pooling them would hide the only thing the experiment can say about them, which is that they behave differently. Splits O1–O6 hold out an entire level of one factor, so the test marginal of that factor lies outside the training range; O7 trains without any disruption and tests under disruption, so both the covariate distribution and the conditional relationship change; O8 is a separate campaign in which the disruption occurs on the north corridor, which carries no disruption in any training context. The taxonomy and the training and test domains are given with the results in Table 7.

For each split the selector, its scales, the fixed-policy baseline, the conformal quantiles and the support envelope are all refitted on the training side, exactly as in the main protocol.

4.7. *Statistical Evaluation*

Every context is replicated with five paired seeds under common random numbers: the vehicle set, departure times, guided and unguided labelling, habitual corridor choice, background traffic and the disruption realisation are functions of the seed and the context only, never of the policy. Two runs of a context under different policies therefore face identical vehicles, and every comparison between policies is a paired comparison.

An advantage of a over b in context s is called *resolved* when, over the paired seed differences $d_i = C_1(s, a, i) - C_1(s, b, i)$,

$$|\bar{d}| > 2\text{sd}(d)/\sqrt{n}. \quad (5)$$

Eq. (5) is a descriptive screen and nothing more: it separates orderings that outrun their own replication scatter from orderings that do not. It is not an inferential procedure and is not offered as one. Five replications leave four degrees of freedom in the standard error, the distribution of the paired differences is not modelled, there is no null hypothesis, and no adjustment is made for making the comparison 648 times. No hypothesis test appears in this paper and no claim of statistical significance is made in it. Differences that fail the criterion are carried through the analysis as ties, and their number is reported alongside every count that depends on them.

Where the mean of a quantity over contexts is reported, its dispersion is characterised by the same paired machinery rather than by an interval derived from an assumed distribution: Section 6.4 compares the headroom in each context against that context’s own noise band, and reports how many contexts carry a headroom larger than it.

A separate convention is needed for ties on the primary criterion, because they are common. In 50 of 648 contexts two or more policies produce *exactly* equal system mean journey time, because at low demand or low penetration they route the guided cohort identically. Winner counts are therefore reported under a declared, stable tie-break — portfolio order $P1 \prec P2 \prec P3 \prec P4$ — alongside the count of contexts with a unique minimum, so that no count depends on an implementation detail of sorting.

5. Experimental Design and Computational Study

5.1. Screening and Study Specification

The specification — factors, levels, policies, parameters, criteria, primary decision cost, resolution criterion, descriptor set, selector ladder, hyperparameters, shift splits and validity rules — was frozen and committed before any evaluation run. Everything reported below follows it, with one recorded exception described in Section 5.7.

Before the main campaign, a 16-context screen with ten seeds per cell (640 runs) evaluated four pre-declared gates on the primary criterion: that at least three policies be strictly best somewhere with a resolved margin (G1); that

at least 25 % of contexts have a non-singleton Pareto set (G2); that policy pairs not be effectively identical on average (G3); and that no single factor explain the winner map almost entirely (G4). G2, G3 and G4 passed. G1 failed, with two policies rather than three.

5.2. *Main Factorial Campaign*

The main campaign is the $648 \times 4 \times 5$ design: every context under every policy at every seed, 12,960 runs. It is the basis of every result in Sections 6.1–6.6 and 6.5.

5.3. *Boundary Refinement*

The main grid samples demand every 600 veh/h, so it can locate a change in the preferred policy only to within a 600 veh/h bracket. Where the resolved winner changes between adjacent demand levels, a refinement campaign inserts three intermediate levels at 150 veh/h spacing inside that bracket, holding all other factors fixed. Nineteen such transitions were refined, at 1140 runs. A grid cannot locate a boundary more finely than its own spacing, so the refined result is reported as a bracket between adjacent sampled levels and never as a continuous threshold.

5.4. *Sensitivity Experiments*

Two policy parameters had to be chosen and were frozen: the admissibility tolerance ϵ of P3 and the risk weight λ of P4. A separate campaign of 1920 runs re-simulates a 96-context subset at $\epsilon \in \{0.1, 0.3\}$ and $\lambda \in \{0.5, 1.5\}$ to measure how much the frozen choice matters. The frozen values are not revised in the light of this experiment; the result is reported as a property of the benchmark (Section 6.9).

A second, purely analytical sensitivity varies the preference profile over the three criteria across three declared normalised weightings, to test whether the choice of preferred policy depends on the weights.

5.5. *Distribution-Shift Experiments*

Splits O1–O7 are constructed from the main campaign by holding out factor levels. O8 required a campaign of its own, because no context in the main factorial contains a disruption anywhere other than on the arterial: 54 contexts with a north-corridor disruption, 1080 runs.

A planned split that proved infeasible. The domain-shift split was originally specified as a disruption on the *south bypass*, which would have been the strongest available test because the bypass is the reliability-aware policy’s preferred corridor. The corresponding campaign of 2160 runs was generated and executed, and every one of the 2160 runs failed. The reason is structural rather than computational: the bypass is a single-lane corridor, so closing a lane on it does not reduce its capacity, it disconnects it, and the simulator correctly reported that affected vehicles had no valid route. A lane closure is a capacity reduction only on a link with more than one lane. The split was replaced by the north-corridor disruption reported as O8, which is a genuine domain shift — no training context contains a disruption on that corridor — but is a weaker test than the one originally planned, because the north corridor is not the corridor that any policy specialises in. The failed campaign is reported in the ledger, the replacement is not presented as the original design, and the substitution is counted as a deviation in Section 5.7.

5.6. Campaign Ledger

Table 3 reconciles every simulation executed in the study. The distinction matters because two different totals are defensible and only one of them supports the results: 12,960 valid runs constitute the main campaign and every result in Sections 6.1–6.6 derives from them, while 20,284 simulations were attempted across all campaigns, of which 18,124 succeeded. Neither number is used where the other is meant.

Table 3: Campaign ledger. Every simulation executed in the study, reconciled. Completion is the fraction of generated vehicles that reached their destination within the simulated horizon; a run is valid only at completion ≥ 0.98 with no teleports.

Campaign	Purpose	Runs	Failed	Contexts	Pol.	Seeds	Compl.
Screen	complementarity screen	640	0	16	4	10	1.000
Validate	scenario validation	384	0	96	4	1	0.973
Main	main factorial	12,960	0	648	4	5	1.000
Boundary	boundary refinement	1,140	0	19 brack.	4	5	1.000
Sensitivity	policy parameters	1,920	0	96	2	5	1.000
O8 north	domain shift	1,080	0	54	4	5	1.000
O8 bypass	domain shift, infeasible	0	2,160	—	4	5	—
Attempted		20,284					
Succeeded		18,124					

5.7. *Experimental Integrity and Deviations*

Two deviations from the frozen specification are recorded, and both are reported here rather than in a supplement because each changes how a reader should weigh a result.

D-1: the screening gate failed and the campaign was run anyway. The frozen rule stated that if G1, G3 or G4 failed, the main campaign would not be run and the study would report a negative result about portfolio design. G1 failed: only two policies, P1 and P3, were strictly best somewhere with a resolved margin, against a threshold of three. The main campaign was run regardless. This is a protocol deviation and is not resolved by reinterpreting the rule.

The reasons were recorded before the campaign was launched and none of them is a result the campaign might produce. First, the gate’s purpose was satisfied: it exists to prevent a selection study being built on a portfolio with no selection problem, and a selection problem demonstrably existed, with the preferred policy reversing between P1 and P3 at resolved margins up to 18.0 %. Second, the gate was written against the primary criterion alone and was blind to the multi-criteria structure beside it: G2 passed, and all four policies appeared in Pareto sets, the reliability-aware policy most often of all. Third, the negative finding was not yet supportable: the screen sampled 16 contexts at extreme factor levels only and never sampled the demand transition region, so stopping would have asserted a negative from a small and badly placed fraction of the design.

Four constraints were declared at the same time, before the campaign: that the study may not claim four-way complementarity on the primary criterion; that it may not claim P2 or P4 is ever the best journey-time policy unless the full campaign resolves such a win; that G1’s failure must be reported as a finding; and that a confirmatory gate G1’ — the identical definition evaluated on all 648 contexts — be declared in advance and its result reported whichever way it came out. Section 6.2 reports that result.

The methodological lesson is recorded as such, because it generalises beyond this study: a complementarity screen must cover the interior of the factor space and not only its corners. A resolution-IV fraction at extreme levels is an efficient design for estimating main effects and a poor one for discovering where a winner map changes. A gate that is passed only after being overridden was a badly designed gate.

D-2: the domain-shift split was replaced. The originally specified O8 was infeasible for the structural reason given in Section 5.5, and was replaced by a weaker but valid alternative. The replacement is labelled as such wherever it appears.

6. Results

6.1. Campaign Validity

All 12,960 runs of the main campaign completed. Completion is 1.0000 in every run, minimum and mean alike, and the total teleport count is zero, so no run was discarded and no context was excluded: all 648 are analysed, each with four policies at five seeds. The campaign contains approximately 22.0 million simulated vehicle trips. Because validity is exact rather than near-exact, none of the results below involves a selection decision about which runs to keep.

6.2. Policy Complementarity

The confirmatory gate. Gate G1', declared before the campaign and evaluated on all 648 contexts, **passes**: three policies — P1, P3 and P4 — are strictly best in at least one context with a resolved margin, against the threshold of three. The resolved winner counts are P3 227, P4 19 and P1 11, with 257 of 648 contexts resolved. **P2 is never a resolved winner**, which confirms on the full factor space what the screen saw in its 16 contexts. The screen's STOP verdict was therefore a sampling artefact of the screening design rather than a property of the portfolio: the screen had the power to detect near-ties, which it did correctly, but not the coverage to find the regions where P1 and P4 win.

Ties are common and are reported as such. In 50 of 648 contexts (7.7%) two or more policies produce exactly equal system mean journey time, because at low demand or low penetration they route the guided cohort identically. The tied sets are P2 with P4 in 39 contexts, P2 with P3 and P4 in 8, and P3 with P4 in 3. Under the declared tie-break the winner counts are P3 418, P4 129, P2 80 and P1 21; restricting to the 598 contexts with a unique minimum gives P3 415, P4 129, P2 33 and P1 21. We report both, because a count that changes by 47 contexts depending on how ties are broken should not be presented as a single number.

Most differences do not survive replication. Of the 648 contexts, 391 (60.3 %) have an unresolved winning margin. Those margins are negligible rather than merely uncertain: their median is 0.13 % of the best journey time, against a median of 5.20 % among resolved margins, which reach 26.9 % (Fig. 2b). With five paired seeds under common random numbers a 5 % difference resolves comfortably, so the unresolved cases are genuine equivalence and not a power failure. The portfolio is complementary, but in most contexts the choice among its members does not matter on journey time.

Two policies are largely interchangeable. Applying the study’s operational-identity criterion — an unresolved paired difference together with a route split differing by less than 2 % of the guided cohort — P2 and P4 are indistinguishable in 42.6 % of contexts, P2 and P3 in 22.7 %, and P3 and P4 in 20.2 %. P1 is never indistinguishable from any other policy, which is expected: it is the only policy that ignores traffic state. Averaged over the six pairs the identity rate is 14.2 %. A nominal portfolio of four policies therefore behaves, over much of the context space, like a portfolio of two or three.

The winner map is multi-factor but shallow. Fig. 2a shows the modal winner over the demand-by-penetration plane. Predicting the winner with a constant is correct in 64.5 % of contexts; the best single factor, demand, reaches 66.5 %; the best pair, demand with penetration, reaches 70.7 %; the best triple, demand with green ratio and penetration, reaches 79.2 %; and all six factors together identify the winner in every context by construction. The structure is genuinely multi-factor — no single factor comes close — but it is also shallow, in the sense that the marginal return to context information falls off quickly after the first two factors. This is the first indication that the decision problem, although real, is not large.

6.3. Mechanisms and Preference Boundaries

The winner map has a physical reading. Fig. 3 shows why the portfolio separates at all: the distance-minimising policy P1 diverges from the other three as demand rises, because it concentrates the guided cohort on the arterial regardless of whether the arterial can carry it, while the three traffic-aware policies remain within a few per cent of each other throughout. The same figure shows the penetration effect that Section 2.1 would predict: mean journey time under the traffic-aware policies falls from penetration 0.2 to 0.5 and then rises again at 0.8, while P1 deteriorates monotonically. More

Table 4: Complementarity of the portfolio over 648 contexts on the primary criterion, and over all three criteria jointly. “Best” uses the declared portfolio-order tie-break; “unique best” counts only the 598 contexts with a unique minimum; “resolved best” requires the margin over the runner-up to exceed twice its paired standard error.

	P1	P2	P3	P4	total
Best on journey time	21	80	418	129	648
of which unique best	21	33	415	129	598
of which resolved best	11	0	227	19	257
Best on CO ₂	10	57	453	128	648
Best on stopped delay	0	80	139	429	648
In the three-criterion Pareto set	21	266	555	525	—
<i>Operationally indistinguishable pairs</i> (share of 648 contexts)					
P2–P4 42.6 %; P2–P3 22.7 %; P3–P4 20.2 %; P1 with any other 0.0 %					
<i>Winner-map classification rate</i>					
constant 64.5 %; best single (demand) 66.5 %; best pair (demand, penetr.) 70.7 %					
best triple (demand, green, penetr.) 79.2 %; all six factors 100 %					

guidance is not monotonically better, consistent with Yang [25] and with the concentration mechanism reported by Cornacchia et al. [6], although this study does not isolate that mechanism and the agreement is a consistency check rather than a replication.

Fitting the depth-2 mechanistic rule to the winner’s identity on all 648 contexts, for exposition, gives a first split on $x_8 = pD/\text{cap}_C$, the share of the shortest corridor’s capacity that the guided cohort alone would consume, at a threshold bracketed between the adjacent sampled values 0.748 and 0.781. Below that threshold the rule separates on the effective green ratio; above it, on whether total demand exceeds arterial capacity. Fig. 4 shows the corresponding structure directly: below $x_8 \approx 0.76$ the reliability-aware and reactive policies win a substantial share of contexts, and above it the load-balancing policy wins almost everywhere.

The reading is that the controlling quantity is not demand and not penetration separately but their product relative to the capacity of the corridor that every policy would otherwise prefer. When the guided cohort alone would saturate the arterial, spreading it across corridors is worth doing; when it would not, spreading it costs more than it saves. That is a statement about association within a designed factorial. The experiment varies the factors but does not isolate the mechanism — it contains no treatment that changes x_8 while holding the physical routes constant — so no causal claim is made, and the rule is reported as consistent with the data rather than as identified by it.

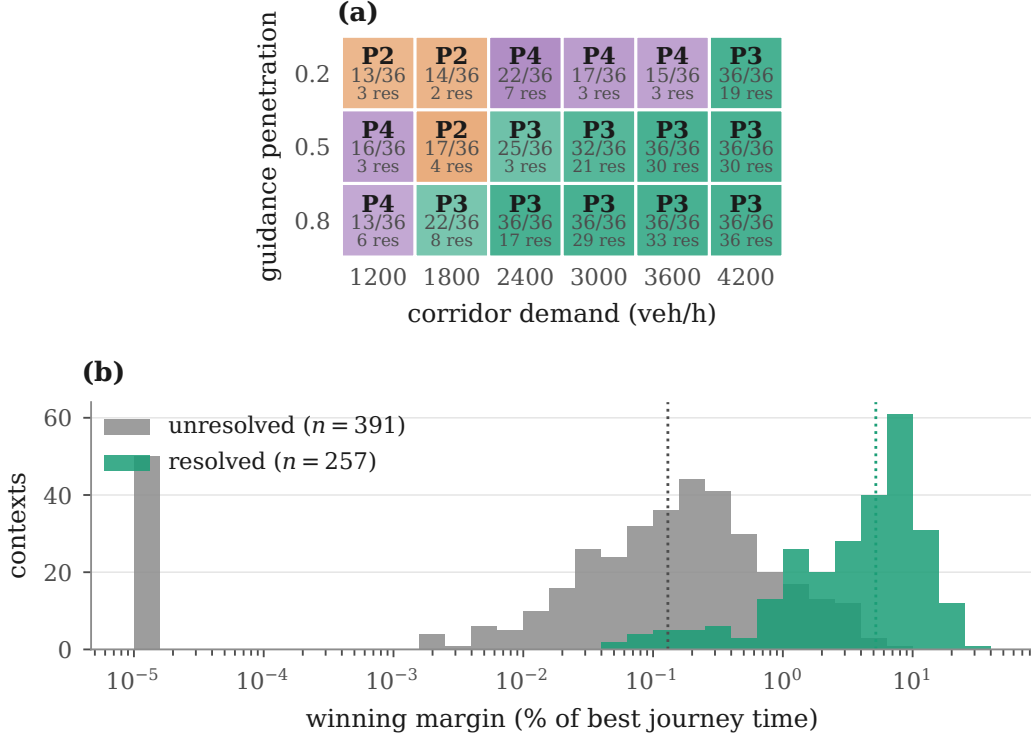


Figure 2: (a) Modal winning policy over the demand-by-penetration plane, each cell aggregating 36 contexts; the fraction is the modal winner’s share of those contexts and “res” counts contexts whose winning margin is resolved against replication noise. (b) Distribution of the winning margin, as a percentage of the best journey time in that context, separated into resolved and unresolved contexts. Dotted lines mark the two medians. The axis is logarithmic and exact ties are shown at the left edge.

Refining the boundary. Nineteen demand transitions were refined from a 600 veh/h grid to a 150 veh/h grid. Thirteen of them have a single changepoint on the refined grid, and for nine of those the point-estimate bracket narrows to 150 veh/h; the remaining four stay at 600. The other six transitions are *not* monotone in demand: the winner changes two to five times across the refined sequence. Moreover, of the thirteen single-changepoint transitions, only seven have either endpoint of the changepoint interval resolved against replication noise. The honest summary is that refinement narrows a point estimate but does not deliver a resolved boundary: a grid cannot locate a boundary more finely than its own spacing, and at this spacing the winner sequence is, in a third of cases, not even monotone. No continuous threshold

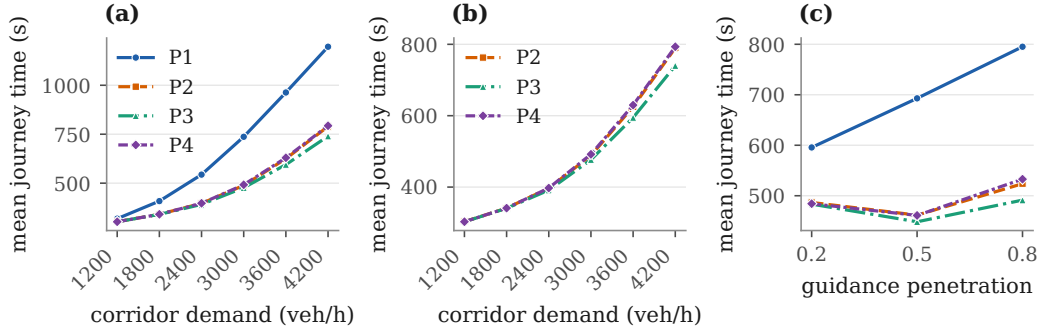


Figure 3: System mean journey time against (a, b) corridor demand and (c) guidance penetration, averaged over all other factors. Panel (b) repeats panel (a) without P1 so that the three traffic-aware policies can be distinguished. Colours and markers are consistent across panels.

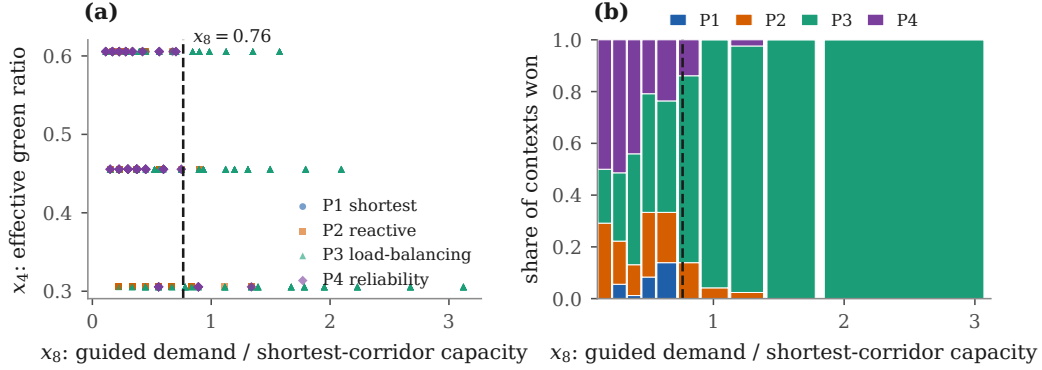


Figure 4: (a) Contexts in the plane of controlled-flow saturation x_8 and effective green ratio x_4 , coloured by the winning policy; the dashed line is the first split of the depth-2 rule fitted to all contexts. (b) Share of contexts won by each policy in deciles of x_8 . The transition is a property of x_8 rather than of demand or penetration separately.

is reported anywhere in this paper.

6.4. Decision Value

This is the central result. The single best fixed policy over all 648 contexts is P3, with a system mean journey time of 474.35 s per vehicle. The hindsight best achieves 473.90 s. The decision headroom is therefore

$$H = 474.35 - 473.90 = 0.45 \text{ s per vehicle}, \quad (6)$$

which is 0.095 % of the fixed policy’s own mean cost. Averaging the per-context ratio instead gives 0.120 %; the two are different statistics and the aggregate is used throughout, with its denominator stated.

Against that, the cost of choosing the wrong fixed policy is large. Table 5 gives the penalties: committing to P1 for the whole benchmark costs 220.21 s per vehicle more than committing to P3, which is 46.4 % of P3’s mean cost. P2 and P4 cost 16.34 s (3.4 %) and 18.34 s (3.9 %). The ratio between the first-order decision — which fixed policy — and the second-order decision — whether to adapt it — is roughly 490 to 1. Fig. 5a places the two on the same axis, which requires a logarithmic scale.

Table 5: Decision value on the primary criterion over 648 contexts. Penalty is the mean cost above the single best fixed policy; headroom is the mean cost of the single best fixed policy above the per-context best.

Choice	Mean journey time (s/veh)	Penalty (s/veh)	Penalty (%)
Fix P1 (shortest)	694.56	220.21	46.42
Fix P4 (reliability)	492.69	18.34	3.87
Fix P2 (reactive)	490.69	16.34	3.44
Fix P3 (load balancing, SBS)	474.35	0.00	—
Hindsight best (VBS)	473.90	−0.45	−0.095

Why the headroom is so small. The fixed policy is near-dominant rather than merely good. P3 attains the per-context minimum in 426 of 648 contexts (65.7 %). In the 222 contexts where it does not, it loses on average 1.32 s and at worst 17.19 s. In the contexts where it does win, it wins by 22.47 s on average. The upside of the fixed policy is therefore about 17 times its downside, and an adaptive layer competes only for the downside. A portfolio

can be complementary in the sense of Section 6.2 while this ratio makes adaptation nearly worthless, and that is what happens here.

Most of the headroom is not resolvable. The headroom must also be compared against the replication noise of the experiment that measured it. Fig. 5c plots, for each of the 222 contexts where the fixed policy is not best, the headroom in that context against that context’s own 2 SE band on the paired difference. Only 55 of the 222 (24.8 %) carry a headroom larger than their own noise band. The mean noise band in those contexts is 1.67 s, against a mean headroom of 1.32 s. Resolving the campaign-wide mean effect of 0.45 s at the same criterion would require approximately 25 seeds per context rather than five, that is about 64,800 runs instead of 12,960. This is stated as a property of the effect size, not as a recommendation: an effect that needs 25 replications per cell to separate from simulation noise is unlikely to be the deciding consideration in a deployment.

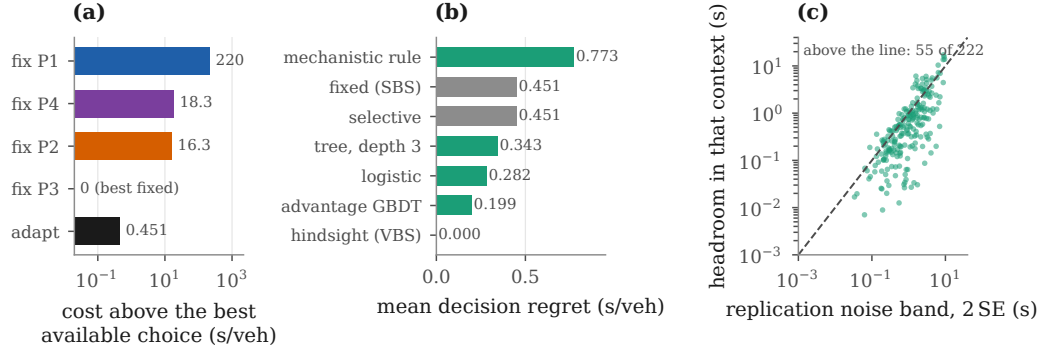


Figure 5: (a) Cost of each available commitment above the best one, on a logarithmic axis: the penalty for the wrong fixed policy dwarfs the headroom available to adaptation. (b) Mean decision regret of every rule under leave-one-context-out evaluation. (c) For each of the 222 contexts where the fixed policy is not the per-context best, the headroom in that context against its own replication noise band; points above the dashed line are contexts whose headroom exceeds the dispersion of the experiment that measured it.

6.5. Selector Comparison

Table 6 reports every rule under identical leave-one-context-out evaluation, in which the model, its scales, the fixed-policy baseline, the conformal quantiles and the support envelope are all refitted from the 647 training contexts of each fold, and all five replications of the held-out context move together. Two findings matter more than the ordering itself.

Table 6: Selector ladder under leave-one-context-out evaluation on 648 contexts. Regret is in seconds per vehicle. “Gap closed” is $G(\pi)$ of Eq. (4): the share of the fixed policy’s headroom recovered. “Top-1” is the fraction of contexts in which the chosen policy is the observed best, and is reported to be contrasted with regret, not to rank the rules.

Rule	Mean	Median	Max	CVaR ₉₀	Top-1	Gap closed
B1 mechanistic rule	0.773	0.000	17.19	6.47	66.7 %	−71.4 %
B0 fixed policy (SBS)	0.451	0.000	17.19	3.67	64.5 %	0.0 %
B5 selective (gated B4)	0.451	0.000	17.19	3.67	64.5 %	0.0 %
B2 tree, depth 3	0.343	0.000	17.19	3.02	73.9 %	+23.8 %
B3 logistic	0.282	0.000	17.19	2.55	76.1 %	+37.4 %
B4 advantage GBDT	0.199	0.000	6.96	1.66	73.1 %	+ 55.8 %
hindsight best (VBS)	0.000	0.000	0.00	0.00	100 %	+100 %

The mechanistic rule is more accurate and more expensive than doing nothing.. B1 identifies the winning policy in 66.7 % of contexts against the fixed policy’s 64.5 %, and yet its mean regret is 0.773 s against 0.451 s — it closes −71 % of the headroom, which is to say it opens a gap of its own. The explanation is visible in the tail: B1 and B0 share the same worst case, but B1’s CVaR₉₀ is 6.47 s against 3.67 s. The rule deviates from the fixed policy in contexts where it is right about the ordering and wrong about the magnitude, gaining a fraction of a second when correct and losing several seconds when not. Selecting the right policy more often is not the same as deciding better, and a study that reported accuracy alone would have concluded that the mechanistic rule works.

Training on decision cost changes the ordering. B3, the most accurate rule at 76.1 %, is not the cheapest. B4 is less accurate at 73.1 % and recovers substantially more headroom, 55.8 % against 37.4 %, because it is the only rule fitted to advantage rather than to the winner’s identity: it learns how much a policy is worth rather than which policy wins, so it deviates from the fixed policy only where the predicted gain is large. The consequence appears in the tail statistics, which are the ones an operator would care about. B4 is the only rule that reduces the worst case, from 17.19 s to 6.96 s, and it more than halves CVaR₉₀, from 3.67 s to 1.66 s. It deviates from the fixed policy in 233 of 648 contexts and leaves the fixed policy in place in the rest. This is the predict-then-optimize distinction [7] appearing in a traffic setting, and it is the one place in this study where the choice of learning target — not the choice of model class — changes the answer.

What the gain is worth. Closing 55.8% of the headroom means reducing mean journey time from 474.35s to 474.10s, a saving of 0.25s per vehicle, or 0.053%. The selector works, in the sense that it does what it was asked to do on held-out contexts, and it is not worth deploying for that gain. Both halves of that sentence are results.

6.6. Multi-Criteria Structure

Complementarity looks quite different on the criterion vector than on the scalar. Over the three criteria jointly, 473 of 648 contexts (73.0%) have more than one non-dominated policy, and the mean Pareto set contains 2.11 of the four policies (Fig. 6a). Pareto membership is P3 in 555 contexts, P4 in 525, P2 in 266 and P1 in 21.

The reliability-aware policy is the clearest case of criterion specialisation. P4 minimises total stopped delay in 429 of 648 contexts, more than any other policy, while minimising journey time in only 129 and CO₂ in 128 (Fig. 6b). It achieves this by routing to the steady unsignalised bypass: vehicles spend longer in the network but less of that time stationary. A study that ranked policies on journey time alone would record P4 as a marginal member of the portfolio; on stopped delay it is the leading one. The distance-minimising policy P1 is the opposite case — it never minimises stopped delay in any context.

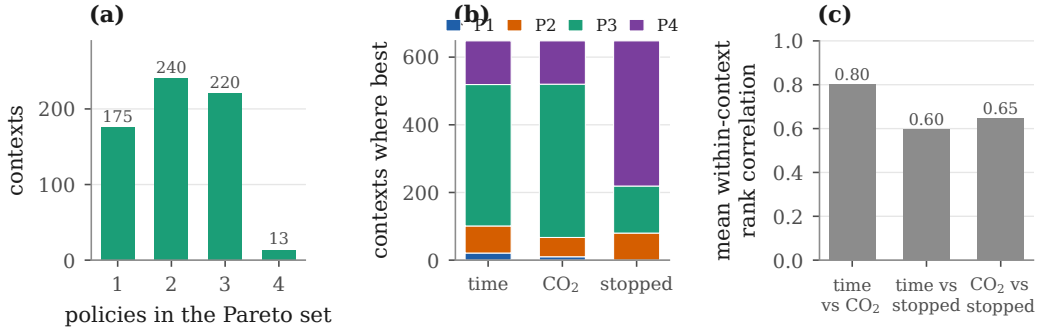


Figure 6: (a) Number of non-dominated policies per context over journey time, CO₂ and stopped delay. (b) Which policy minimises each criterion, counted over the 648 contexts. (c) Mean within-context Spearman correlation between criteria across the four policies, which is the correlation that governs whether the criteria order policies the same way.

The criteria are correlated but not interchangeable. Pooled across all context-policy cells the Pearson correlations are 0.94 for journey time against

CO₂, 0.89 against stopped delay and 0.89 between CO₂ and stopped delay; but the quantity that matters for a decision is the within-context correlation across the four policies, and that is 0.80, 0.60 and 0.65 respectively (Fig. 6c). The best policy on CO₂ differs from the best on journey time in 192 contexts (29.6 %), and the best on stopped delay differs in 363 (56.0 %).

Two consequences follow. First, stopped delay carries decision information that journey time and CO₂ do not, and it is the criterion on which the portfolio is most complementary. Second, the close alignment of journey time and CO₂ is the evidence that led to eco-routing being excluded from the portfolio: with a within-context rank correlation of 0.80 and identical policy orderings in 52.2 % of contexts, a standalone environmental objective has little room to differ from a time-based one in this network. That exclusion is therefore a finding about this environment, not an assumption, and it should not be read as a general statement about eco-routing.

Crucially, a large Pareto set is not evidence that adaptation is valuable. A set can be large because the criteria disagree about an ordering while all the differences are small. That is what happens here: 73.0 % of contexts have more than one non-dominated policy, and the decision value on the primary criterion is 0.45s per vehicle. Complementarity on the criterion vector and decision value on the decision criterion are different quantities, and this benchmark separates them by two orders of magnitude.

6.7. Distribution Shift

Table 7 reports the eight splits separately. Pooling them would produce an average that describes none of them.

Table 7: Distribution shift, by type. Regret is mean decision regret in seconds per vehicle on the held-out contexts. “Flagged” is the share of held-out contexts the support gate declared outside the training envelope; “abstained” is the share on which the selective rule declined to act. A split is classified as helping or harming when the selector’s regret differs from the fixed policy’s by more than the declared 0.05s tolerance.

Split	Shift type	Tr.	Te.	Fixed	GBDT	Selec.	Flag.	Verdict
O3 demand = 2400	interpolation (control)	540	108	0.565	0.167	0.565	0 %	helps
O2 demand = 1200	extrapolation, below	540	108	0.528	0.531	0.528	15 %	neutral
O1 demand = 4200	extrapolation, above	540	108	0.000	0.983	0.000	43 %	harms
O4 lag = 300 s	extrapolation, above	432	216	0.232	0.237	0.232	100 %	neutral
O5 penetration = 0.8	extrapolation, above	432	216	0.298	0.293	0.298	100 %	neutral
O6 green = 0.65	extrapolation, above	432	216	1.040	1.157	1.040	100 %	harms
O7 disruption present	covariate + concept	324	324	0.630	2.216	0.630	32 %	harms
O8 north disruption	domain shift	648	54	0.314	0.176	0.314	0 %	helps

At the declared tolerance the unguarded selector **helps on two** splits (the interpolation control O3 and the domain shift O8), is **indistinguishable on three** (O2, O4, O5) and **harms on three** (O1, O6 and, worst, the concept shift O7). It is not the case that the selector degrades under every shift, and it is not the case that shift is harmless; the behaviour depends on the shift, which is why the splits are not pooled.

Three observations carry most of the content.

The worst failure is the concept shift. O7 trains without any disruption and tests under disruption, so both the covariate distribution and the conditional relationship between context and policy performance change. The fixed policy incurs 0.630s of regret there; the selector incurs 2.216s, more than three times as much. Extrapolating a factor level is survivable in this benchmark; learning a relationship that no longer holds is not.

Extrapolation above the training range is where the selector does damage.. O1 holds out the highest demand. There the fixed policy is optimal in every one of the 108 held-out contexts, so its regret is exactly zero, and any deviation is pure loss: the selector incurs 0.983s. A tree-based regressor cannot extrapolate — beyond the largest training value of a feature it returns the prediction of its outermost leaf — and here that leaf was fitted where the fixed policy was not always best.

The support gate does not detect the shift it was built for. The gate flags 100 % of held-out contexts on the three extrapolation splits O4, O5 and O6, which is the behaviour it was designed for. It flags 43 % on O1 and 32 % on O7. It flags **none** of the 54 held-out contexts of O8 (Fig. 7b), despite O8 being the only genuine domain shift in the set. The reason is not a tuning failure but a representation failure: the descriptor x_7 encodes that a disruption may occur, not where it occurs, so a disruption relocated to a different corridor is invisible in the feature space the gate operates in. A detector that measures distance in a feature space cannot detect a change that the feature space does not encode. This is a general limitation of support-based out-of-distribution detection, and it is exactly the case in which such a gate would be most relied upon. No claim of out-of- distribution safety is made anywhere in this paper.

6.8. Selective Prediction

The selective rule B5 acts only when the context is inside the support envelope and the split-conformal lower bound on the predicted advantage is

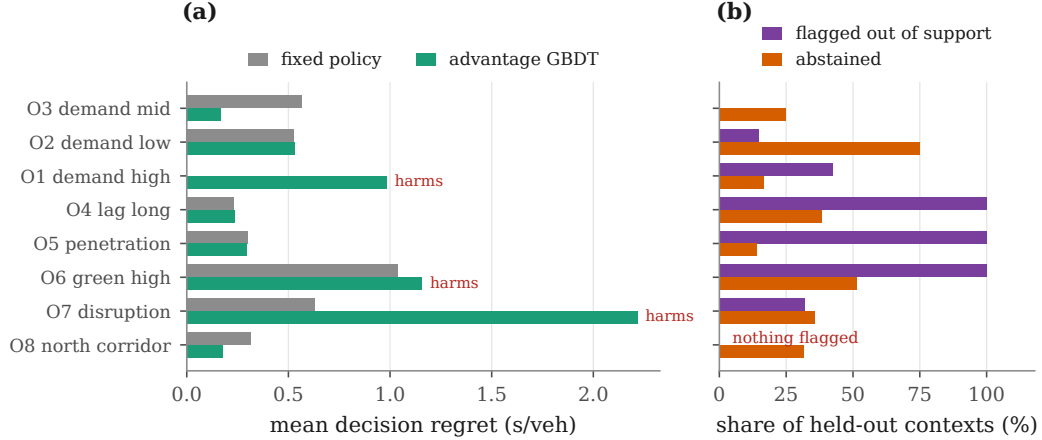


Figure 7: (a) Mean decision regret of the fixed policy and the advantage GBDT on each held-out split. (b) Share of held-out contexts flagged outside the training support and share on which the selective rule abstained. The support gate flags every context of the three factor-level extrapolations and none of the domain shift O8.

positive. Its behaviour in this study is unambiguous and entirely negative:

B5 chose the fixed policy in *every one* of the 648 contexts, and on *every one* of the eight shift splits. It abstained in exactly the 233 contexts in which the unguarded selector would have deviated, and in the remaining 415 the unguarded selector chose the fixed policy anyway. The confidence gate never certified a single deviation.

This is not a tuning failure, and it is not a property of the particular conformal construction. It follows from a ratio. The mean replication noise floor across contexts — the largest paired 2 SE against the best policy — is 20.90 s per vehicle. The mean advantage available to any deviation is 0.45 s. A conformal interval calibrated to residuals of that magnitude is necessarily far wider than the effect it is being asked to certify, so the lower bound on a predicted advantage is essentially never positive. Any confidence-gated selector, conformal or otherwise, faces the same arithmetic in this regime.

The consequence is a design rule that generalises beyond this study, and it is the reverse of the usual framing. Selective prediction is normally presented as insurance: give up a little in-distribution performance to avoid out-of-distribution harm. Here the insurance costs everything and pays nothing, because the premium — the entire 55.8 % of headroom that B4 recovers — is

larger than the loss being insured against in every split where the selector helps, and the gate cannot distinguish the splits where it harms. Before building an abstention mechanism, the effect size should be compared against the replication noise of the system it will act on; if the ratio is of the order seen here, no confidence gate can act.

6.9. Sensitivity and Robustness

Policy parameters. Two parameters had to be chosen and were frozen: the admissibility tolerance ϵ of P3 and the risk weight λ of P4. Re-simulating a 96-context subset shows that the choice matters, and in one case matters a great deal. Loosening P3’s tolerance from the frozen $\epsilon = 0.20$ to 0.30 improves its mean journey time by 16.66 s per vehicle; tightening it to 0.10 costs 8.10 s. Changing P4’s λ from 1.0 to 0.5 or 1.5 changes its mean cost by -0.11 s and $+1.27$ s, with a median change of exactly zero in both cases, so P4 is insensitive to its parameter over this range while P3 is not.

The frozen values are not revised, and the main analysis is not recomputed at the better value: doing so after seeing the result would replace a pre-declared benchmark with a tuned one. The correct reading is that the benchmark is *conservative* with respect to P3 and that parameter tuning within a single policy is worth roughly 37 times more on this benchmark than selecting among policies. That comparison is arguably the most practically useful number in the study, and it was produced by a robustness check rather than by the main experiment.

Preference weights. Scoring the three criteria under three declared normalised profiles changes the preferred policy in 14.5 % of contexts. The time-weighted profile agrees with the primary criterion in 92.9 % of contexts, the balanced profile in 86.0 % and the environment-weighted profile in 80.1 %. The decision is therefore not inert with respect to the preference model — in contrast to what a strongly aligned criterion set would produce — but neither is it dominated by it.

Model capacity. The ladder of Table 6 is itself the capacity sensitivity: depth-2 tree, depth-3 tree, logistic regression and gradient boosting span a wide range of flexibility, and the ordering by regret does not follow the ordering by capacity. The best rule by regret is not the best by accuracy, and the simplest rule is worse than no rule at all.

7. Discussion

7.1. What the Experiment Establishes

Three things are established on this benchmark, and they should be stated with their scope attached. First, the portfolio is complementary: three of four policies are strictly best somewhere with a margin that survives replication, and 73.0% of contexts have more than one non-dominated policy on the criterion vector. Second, the decision value of that complementarity is 0.45 s per vehicle on a 474.4 s mean, against 220.2 s for the first-order choice of which fixed policy to adopt. Third, a selector trained on decision cost recovers 55.8% of the headroom on held-out contexts and reduces the worst case by more than half, while remaining worth 0.25 s per vehicle.

What is *not* established is anything about a real network. One synthetic three-corridor topology, one origin–destination structure, one signal plan family and one demand profile were studied. The numbers above are properties of that benchmark.

7.2. Why Complementarity Does Not Imply Decision Value

The gap between the second and third paragraphs above is the paper’s main point, and it has a mechanism rather than being an accident of this environment.

Complementarity is a statement about the *argmin* of a cost function over a portfolio; decision value is a statement about the *gap* between the best fixed member and that argmin. The two coincide only when the best fixed policy is badly wrong where it is wrong. Here it is not: P3 attains the minimum in two thirds of contexts, and in the third where it does not it loses 1.32 s on average while winning by 22.47 s where it does. An upside-to-downside ratio of 17 means the fixed policy absorbs most of the variation in the winner map, leaving adaptation to compete for a residue.

This has a practical reading that is easy to miss. The right question to ask of a portfolio is not “do different members win in different instances?” — in any non-degenerate portfolio some will — but “how much does the best fixed member lose where it loses, and how often?” Those two quantities, not the count of distinct winners, determine whether per-instance selection is worth building. They are cheap to compute from the same data that establishes complementarity, and reporting only the first while implying the second overstates the case for adaptation.

The multi-criteria result reinforces the same point from the other side. On the criterion vector the portfolio looks strongly complementary — 73.0 % of contexts admit more than one non-dominated policy, and the reliability-aware policy is the best stopped-delay policy in two thirds of contexts while almost never being the best journey-time policy. None of that changes the decision value on the decision criterion. A Pareto set can be large because criteria disagree about an ordering while every difference is small.

7.3. Mechanistic Reading of the Boundary

Where the preferred policy does change, the organising quantity is the product of demand and penetration relative to the capacity of the corridor every policy would otherwise prefer. Below a controlled-flow saturation of roughly 0.76 the guided cohort is small enough, or the arterial large enough, that diverting traffic costs more than the congestion it avoids, and the reliability-aware and reactive policies hold much of the space. Above it the arterial cannot absorb the guided cohort and load balancing wins almost everywhere. The quantity is dimensionless and computable before any vehicle departs, which is what makes it usable as a switching criterion.

Three qualifications belong with that reading. The threshold is a bracket between adjacent sampled levels, not a number: a grid cannot locate a boundary more finely than its own spacing, and the refinement campaign showed that at 150 veh/h spacing a third of the transitions are not even monotone in demand. The rule is an association within a designed factorial; the experiment contains no treatment that manipulates x_8 while holding the physical routes fixed, so it does not identify a mechanism. And the same rule, evaluated as a decision rule rather than as a description, is worse than doing nothing — which is the subject of the next subsection.

7.4. Why Decision-Oriented Evaluation Changes the Answer

The mechanistic rule is more accurate than the fixed policy and costs more than it. The most accurate learned rule is not the cheapest. The cheapest rule is the only one trained on advantage rather than on the winner’s identity. Taken together these say that in this problem accuracy and decision quality are not merely imperfectly correlated; they order the candidate rules differently.

The reason is structural rather than incidental. The distribution of regret is extremely skewed: in most contexts every policy is within a fraction of a second of every other, and in a small minority the gap is tens of seconds. A

rule fitted to the winner’s label spends its capacity on the many contexts where the label is almost arbitrary, and pays for its errors in the few where it is not. A rule fitted to advantage sees the skew directly: it is indifferent where the advantage is small, which is most places, and deviates only where it is large. That is why B4 alone improves the tail — worst case from 17.19 s to 6.96 s, CVaR₉₀ from 3.67 s to 1.66 s — and why it does so while being *less* accurate than the rule it beats.

For practice this means that selection accuracy is the wrong acceptance criterion for a routing-policy selector, and that a selector should be trained on the cost of its decisions when those costs are available, which in a simulation study they always are. For this literature it means that reported accuracies of instance-selection systems are not comparable across problems whose regret distributions differ in skew.

7.5. Distribution Shift as a Deployment Problem

The shift results are not a benchmark of a learning method; they are a statement about what an operator would face. Three points follow from them.

Shift is not one thing. The same selector helps on an interpolation control and a domain shift, is indistinguishable on three factor-level extrapolations, and harms on two extrapolations and a concept shift. Reporting a single out-of-distribution number would have described none of these cases, and would have hidden the one that matters most: the concept shift, where the relationship between context and policy performance itself changes, is where the damage is worst, at more than three times the fixed policy’s regret.

Support-based detection is bounded by its representation. The gate flagged every held-out context of the three factor-level extrapolations — exactly what it was built to do — and none of the domain shift, because the descriptor set encodes whether a disruption may occur but not where. This is not a defect of the particular gate. Any detector that measures distance in a feature space is blind to a change the feature space does not encode, and the changes an operator most needs to catch are often of that kind, because they are the ones nobody anticipated when the features were designed. It follows that an out-of-distribution guard should be specified together with an explicit list of the changes it can and cannot see, and that this list is a property of the features and not of the detector.

The failed bypass-disruption campaign is relevant here rather than merely procedural. The strongest available domain-shift test could not be executed

because closing a lane on a single-lane corridor disconnects it instead of constraining it. The replacement test is real but weaker, and the consequence is that the study’s evidence on domain shift rests on the corridor that no policy specialises in.

7.6. Selective Prediction and the Effect-to-Noise Ratio

The selective rule never acted. That result is worth more than a working gate would have been, because it identifies a precondition that is rarely checked.

A confidence-gated selector certifies a deviation when a lower bound on the predicted advantage is positive. The width of that bound is set by the residual dispersion of the prediction problem, which here is governed by the replication noise of the simulation: a mean 2 SE noise floor of 20.90 s against a mean advantage of 0.45 s. No calibration of the interval can make it narrower than the noise it is estimated from, so the gate cannot certify anything, and the in-distribution gain is forgone in full.

The design rule is therefore to compute the ratio of the available effect to the replication noise *before* building an abstention mechanism. Where that ratio is of the order seen here, a confidence gate is not conservative; it is inert, and the honest options are to accept the unguarded selector with its shift behaviour documented, to raise the effect size by changing the portfolio, or to reduce the noise by replication — which here would require about 25 seeds per context, five times the campaign that was run.

7.7. Implications for Routing-System Design

The results support a small number of concrete statements, all scoped to this environment.

The first-order decision is which fixed policy to adopt, and it is worth roughly 490 times the second-order decision of whether to adapt it. An operator with a limited engineering budget should spend it on that choice, on the calibration of the chosen policy’s parameters — where loosening one tolerance was worth 37 times the entire adaptation headroom — and on measurement, rather than on an adaptive layer.

If an adaptive layer is nonetheless built, three things follow from the results. It should be trained on decision cost rather than on the winner’s identity, because that is what separates the rule that helps from the rule that hurts. It should be given an explicitly declared operating domain, because the harm under shift is concentrated in extrapolation above the training range

and in concept change. And its guard should not be assumed to detect shifts that its features do not encode.

Finally, the reliability-aware policy is worth retaining even though it almost never minimises journey time, because it minimises stopped delay in two thirds of contexts. Which of those an operator wants is a policy question and not a technical one, but the choice should be made explicitly rather than inherited from whichever criterion the evaluation happened to scalarise.

7.8. *Limitations and Threats to Validity*

External validity. One synthetic network with three parallel routes, one origin–destination structure, a homogeneous fleet, and specified rather than field-calibrated demand and signal plans. Corridor capacities were measured within the simulation, not from field data. Nothing here establishes the numerical values for a real network, and the three-corridor structure in particular makes the candidate path set complete, which is a simplification that removes the path-generation problem entirely. *This does not invalidate* the separation between complementarity, decision value and deployability, which is a relationship between quantities rather than a value of any of them, but it does mean the magnitudes are benchmark-specific.

Behavioural assumptions. Guided vehicles comply fully with the policy; unguided vehicles follow a fixed habitual choice; routing is pre-trip with no en-route re-routing; there is no day-to-day learning and no response to the guidance system’s own past behaviour. Real compliance is partial and adaptive [21], and a system that changes its policy would face travellers who respond to that change. *This does not invalidate* the within-benchmark comparisons, which are paired and hold behaviour fixed across policies, but it does mean the penetration results describe an idealised cohort.

Portfolio scope. Four policies, with parameters frozen before the study. The sensitivity experiment shows the frozen tolerance of P3 to be conservative by 16.66 s per vehicle, so the benchmark understates what that policy family can do, and a differently tuned portfolio would have a different headroom. Eco-routing was excluded on screening evidence specific to this network. *This does not invalidate* the reported headroom, which is a property of the portfolio as specified, but it does mean the headroom is not an upper bound over all portfolios.

Statistical resolution. Five paired seeds per context. Under the declared criterion 60.3 % of winning margins are unresolved, and only 24.8 % of the contexts carrying headroom carry more than their own noise band. Resolving the campaign-wide mean would need about 25 seeds. *This does not invalidate* the resolved findings — and the resolved margins are large, with a median of 5.2 % — but it does mean that fine structure in the winner map is beyond this experiment’s resolution, which is why boundaries are reported as brackets.

Simulation and measurement. Emissions come from a model, not measurement, and the three criteria are computed from one simulator’s outputs. Journey time includes insertion delay, which is the right convention for a policy that can oversaturate the network but makes absolute values at the highest demand partly a statement about the insertion queue. *This does not invalidate* the policy ordering, which is unaffected by the accounting boundary, but absolute values should not be compared across studies.

Protocol deviations. Two are recorded in Section 5.7: the main campaign was run after a pre-declared stop gate failed, and the domain-shift split was replaced after the original proved infeasible. The first is mitigated by the confirmatory gate declared in advance and by the constraints adopted at the same time, all of which the results honour; it nonetheless means the study proceeded past a rule it had set itself. The second means the domain-shift evidence is weaker than designed. *Neither invalidates* a reported result, and both are stated so that a reader can discount them as they see fit.

Causal identification. The experiment is a factorial over context variables, not an intervention on mechanisms. Every mechanistic statement in Section 7.3 is an association consistent with the data and is labelled as such.

7.9. Research Implications

Three lines follow from this study rather than being asserted by it. The first is whether the complementarity–value gap survives in networks whose path sets are not complete and whose alternatives are less symmetric; the mechanism identified here — a near-dominant fixed policy with a large upside-to-downside ratio — is not obviously special to three corridors, but nothing here shows that. The second is whether decision-cost training retains its advantage when the regret distribution is less skewed, since the argument for it given in Section 7.4 depends on that skew. The third is the design of out-of-distribution guards whose coverage is stated in terms of what the

feature representation encodes, which the O8 result suggests is the binding constraint rather than the calibration method.

8. Conclusion

This paper treated the choice among established urban routing policies as a per-instance decision problem and measured it on a controlled factorial of 648 operating contexts, four policies and five paired seeds, comprising 12,960 valid microsimulation runs with no failures and no teleports.

The portfolio is complementary. Three of four policies are strictly best in at least one context with a margin that survives replication, the reactive travel-time policy is never a resolved winner, and 73.0 % of contexts have more than one non-dominated policy across journey time, CO₂ and stopped delay. The winner map is genuinely multi-factor: no single factor predicts it above 66.5 %, and the organising quantity is the product of demand and guidance penetration relative to the capacity of the shortest corridor.

The decision value of that complementarity is small. The single best fixed policy leaves 0.45 s per vehicle of headroom on a mean journey time of 474.4 s, because it attains the per-context minimum in two thirds of contexts and, where it does not, loses 1.32 s on average against winning by 22.47 s where it does. By contrast, committing to the worst fixed policy costs 220.2 s per vehicle, 46.4 % of the best fixed policy’s mean. The first-order choice is worth roughly 490 times the second-order one, and tuning a single policy parameter was worth 37 times the entire adaptation headroom.

Deployability is a third and separate question. Under leave-one-context-out evaluation a gradient-boosted model trained on advantage recovers 55.8 % of the headroom and more than halves the worst case, while a transparent depth-2 rule is more accurate than doing nothing and costs 71 % more; accuracy and decision quality order the candidate rules differently. Across eight distribution shifts reported separately by type, the selector helps on two, is indistinguishable on three and harms on three, with the worst damage under concept shift. A support-and-confidence gate never certified a deviation in any of the 648 contexts, because the conformal interval is two orders of magnitude wider than the effect, and its support component flagged every context of three factor-level extrapolations but none of the domain shift, which its feature representation cannot see.

The contribution is therefore a measurement and a set of distinctions, not an algorithm. Complementarity, decision value and deployability are different

properties; they are quantitatively disconnected in this environment; and a study that reports the first as evidence for the third — as is common when portfolios are described by how many members win somewhere — would overstate the case for adaptive routing-policy selection by more than two orders of magnitude. The practical recommendation that follows is correspondingly modest: choose the fixed policy carefully, calibrate it, and add an adaptive layer only after measuring that the headroom it competes for exceeds the noise of the system it will act on.

All results are scoped to one synthetic network and are not externally validated. The specification, run records, analysis code and figure scripts are available so that every number can be recomputed from the simulation output.

Data and code availability

The frozen specification, all run records, the analysis and figure scripts and the verification harness are provided with the submission, so that every number in this paper can be recomputed from the simulation output. Reproduction instructions and software versions are given in the accompanying reproducibility note.

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